

Early Usability Evaluation of a Relational Agent for the COVID-19 Pandemic

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ABSTRACT

Relational agents (RAs) have shown effectiveness in various health interventions with and without healthcare professionals (HCPs) and hospital facilities. RAs have not been widely researched in COVID-19 context, although they can give health interventions during the pandemic. Addressing this gap, this work presents an early usability evaluation of a prototypical RA, which is iteratively designed and developed in collaboration with infected patients (n=21) and two groups of HCPs (n=19, n=16) to aid COVID-19 patients at various stages about four main tasks: testing guidance, support during self-isolation, handling emergency situations, and promoting post-infection mental well-being. The prototype obtained an average score of 58.82 on the system usability scale (SUS) after being evaluated by 98 people. This result implies that the suggested design still needs to be improved for greater usability and adoption.

CCS CONCEPTS

• **Human-centered computing** → **Usability testing**.

KEYWORDS

COVID-19, relational agent, mHealth, healthcare

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1 INTRODUCTION

A relational agent (RA), a special type of a ‘conversational agent (CA)’, is an artificial intelligence (AI) based computational artifact that is responsible for maintaining a virtual socio-emotional relationship with the user for a long time [3]. RAs can be embodied with empathy allowing them to use both verbal and nonverbal relational affects over an extended time to form long-term, deep, and meaningful connections with the users. RAs have been used for different healthcare services such as screening, counseling and caregiving [2, 7, 9] in diverse healthcare settings. Since similar healthcare services are essential for COVID-19 patients [8, 11, 12], RAs can also be used to support the infected individuals as RAs can

remotely deliver essential health services [10], minimizing face-to-face interactions and preventing the transmission of infection. The goal of this research was to design an RA-based intervention that can deliver healthcare services to patients in non-life-threatening situations as they go through different stages of COVID-19. This work presents an early usability evaluation of a prototypical RA that was designed for COVID-19 patients using a user-centered design (UCD) approach.

2 DESIGN REQUIREMENTS

This research has been approved by the institutional review board of our home institution.

User Personas: The intended user groups for the RA were identified through a review of CDC’s COVID-19 prevention guidelines [4] and other related manuscripts published in academic journals. An interview with 19 healthcare professionals (HCPs) who had served COVID-19 patients during the pandemic was later conducted to confirm initial findings [6] which led to the identification of three user groups (personas). Individuals who have some known COVID-19 symptoms (fever, cough, etc.) and seeking testing guidance make up the first persona. The second persona includes individuals who have been diagnosed with mild COVID-19 symptoms and are in home isolation. The third persona deals with the recovered patients who have just returned home from the hospitals and need support for their mental well-being.

Task Scenarios : To identify the intended goals of the RA to serve target users, we conducted an online survey with 21 individuals who had been through the COVID-19 infection and recovery stages [6]. The participants enumerated challenges and difficulties they faced during different stages of the disease as well as the support they required to overcome those challenges. The analysis of the survey responses led to identifying four distinct tasks that the proposed RA needed to design for- (i) Testing Guidance: RA provides testing guidance to a user who suspects getting infected by periodically engaging in a dialogue to obtain up-to-date symptom status and health metrics; (ii) Wellness Support: RA provides wellness tips and companionship to the user during home-isolation; (iii) Handling Emergencies: RA monitors user’s symptoms to avert and prevent emergencies. In such cases, the RA takes appropriate steps to detect critical situations and connects the user to emergency services of a nearby hospital; and (iv) Post-infection Care: RA provides companionship and mental health counseling to help recover from the stress of the infection during the recovery phase.

Conversation Design: Dialogue scripts for each task scenario were prepared in consultation with 16 HCPs [5]. Sample scripts were first prepared and then sent to the HCPs for feedback via surveys. Based on thematic analysis, we identified four major characteristics

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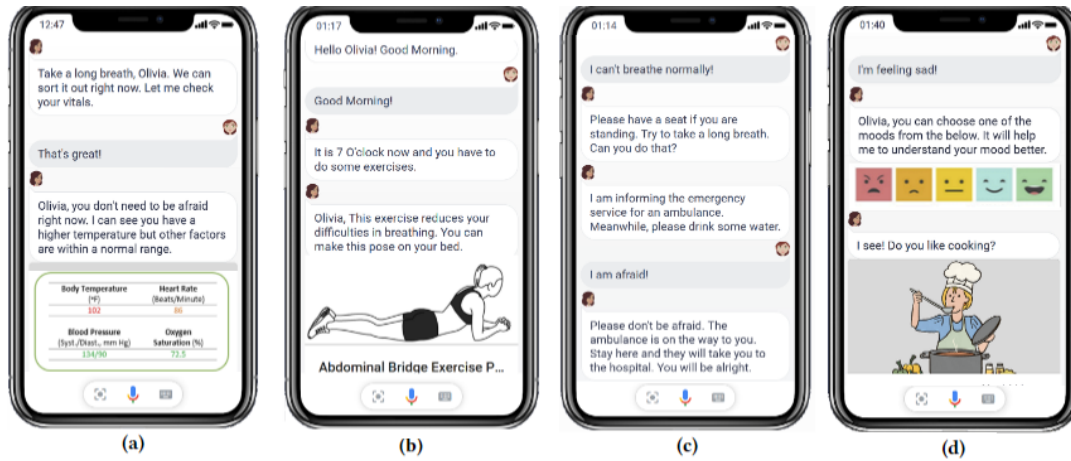


Figure 1: Snippets from the prototype interface (smartphone view). (a) RA checks and shows physiological vitals and diagnoses for any COVID-19 symptoms; (b) RA helps maintain healthy practices for fast recovery during home-isolation; (c) RA takes action during an emergency condition and tries to make user confident by providing affirmative responses; and (d) RA attempts to engage user in daily activities for improving mental well-being.

required for the dialogue scripts: (i) robust and validated screening, (ii) focus on building trust by validating feelings and setting realistic expectations, and (iii) long-term relationship building by frequent check-ins, peer support, and psychological interventions.

3 PROTOTYPE IMPLEMENTATION

Scripts from the Conversation Design phase were used to program the dialogue between the RA and the users. The prototypical RA was developed using a web-based real-time service called *BotSociety* (www.botsociety.io), which utilizes the XML-based Speech Synthesis Markup Language (SSML) to process natural language. With *BotSociety*, the prototype can be distributed for testing and evaluation purposes via a hyperlink, allowing it to be used on any web browser. This flexibility makes it possible to test user experience on different devices.

The RA provides verbal and visual (prompts, graphics) information to the user. The user inputs data using voice or touch/clicks. For improved understanding of the communication between the RA and the user, all voice interactions are also displayed as text on the RA's interface. The user can choose to run the prototype on any of the three device modes, i.e., smartphone, voice assistant, and tablet computer. The RA elicits the user's preferences about the intended scenario in the beginning to generate customized recommendations. The user's COVID-19 infection status was checked at the beginning of each interaction. Fig. 1 illustrates a few snippets from the presented prototype.

4 PRELIMINARY USABILITY EVALUATION

This study was conducted remotely as a user survey targeting infected and recovered individuals so that all the features or scenarios could be evaluated based on participants' infection experience and post-recovery constraints. The survey was designed in Google Forms containing links to the prototype and the task order was controlled. The survey link was posted on various social media

platforms, e.g., Facebook and mailing lists. The recruitment had no geographical constraints. The inclusion criteria were: (a) at least 18 years old, (b) previously or currently infected with COVID-19, and (c) basic understanding of English. The participation was voluntary and no identifying information was collected. 98 (male=53) eligible individuals aged between 18 and 64 years (mean=34.42 years) completed the survey. 52 participants were infected (Symptoms: severe=25, mild=27) and 46 had recovered from the infection (Symptoms: severe=17, mild=29) during the survey. The survey took 45–60 minutes to complete and all the participants interacted with the RA through all the available modalities (voice, text, click/touch, prompts).

Participants completed four study tasks corresponding to each interaction task scenario. Each study task involved, first, reading the patient persona description and then interacting with the prototype using provided prompts. After completing entire tasks, participants completed the System Usability Scale (SUS) [1] to determine the usability of the prototype. The overall average SUS score was 58.82. The average SUS score (59.88) of mildly infected participants was slightly higher than that (57.75) of severely infected participants. However, the difference between the scores of both groups was not statistically significant according to the student T-Test ($P > 0.05$).

5 LIMITATION AND CONCLUSION

The study is characterized by an imbalanced ratio between people with mild and severe symptoms. Even though participants with severe symptoms participated in the study, their health conditions, e.g., weakness, lack of concentration, etc., may have prevented them from responding properly during the study. The interactions with the RA were canned and limited, which may not be a very natural interaction for participants. However, the work is the beginning of an intervention that has the potential to serve people during a COVID-19-like pandemic and a high-fidelity prototype is under development for in-situ evaluation with the target population.

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